

Exhibit A

**STATE OF ILLINOIS
ILLINOIS COMMERCE COMMISSION**

Commonwealth Edison Company	)	
	)	Docket No. 25-0677
Filing proposing changes to new service requests	)	
for large demand project applicants or customers.	)	(consolidated)
(tariffs filed June 23, 2025)	)	
	)	
Filing proposing revisions to Rider Distributed	)	Docket No. 25-0679
System Extensions. (tariffs filed June 23, 2025)	)	

DIRECT TESTIMONY OF

KYLE THOMAS

ON BEHALF OF

**THE ENVIRONMENTAL LAW & POLICY CENTER, NATURAL RESOURCES
DEFENSE COUNCIL, UNION OF CONCERNED SCIENTISTS, AND VOTE SOLAR
("JOINT NGOs")**

September 26, 2025

TABLE OF CONTENTS

I.	WITNESS IDENTIFICATION	1
II.	COMED’S PROPOSED TARIFF REVISIONS	4
III.	COMED'S 50 MW THRESHOLD FOR LARGE LOAD PROVISIONS	5
IV.	TECHNICAL CONCERNS WITH LARGE LOAD INTERCONNECTIONS.....	6
V.	CONCLUSION AND OVERALL RECOMMENDATIONS.....	16

1 **I. WITNESS IDENTIFICATION**

2 **Q: Please state your name, title, and position.**

3 A: My name is Kyle Thomas, Vice President of Engineering and Compliance Services at
4 Elevate Energy Consulting, Inc.

5 **Q: For whom are you testifying in this matter?**

6 A: I am testifying on behalf of the Environmental Law & Policy Center (“ELPC”), Natural
7 Resources Defense Council (“NRDC”), Union of Concerned Scientists (“UCS”), and Vote
8 Solar, together the Joint Non-Governmental Organizations (“Joint NGOs”).

9 **Q: Please describe your formal education and professional experience.**

10 A: I earned a Bachelor of Science and Master of Science in Electrical Engineering from
11 Virginia Tech. Before assuming my current role at Elevate Energy Consulting, I served as
12 Director of Product at VEIR from 2022 to 2024, a startup company developing high-
13 temperature superconducting transmission and distribution lines with innovative new
14 cooling systems. Prior to that, I served as Manager of Electric Transmission Planning and
15 Strategic Initiatives at Dominion Energy Virginia, where I was responsible for the team
16 doing long-term transmission planning for Dominion’s electric transmission system
17 covering most of the state of Virginia and northeast North Carolina, with goals of
18 interconnecting a rapidly growing queue of generation resources, including inverter-based
19 resources, and data centers—primarily located up in the Northern Virginia region of the
20 state. I began my career at Dominion Energy Virginia in 2009, and throughout my career
21 there I held many engineering and leadership roles in the electric transmission business
22 unit. I am also a Licensed Professional Engineer in the Commonwealth of Virginia. My
23 resume is attached as JNGO Ex. 2.01.

24 **Q: Have you previously testified before the Illinois Commerce Commission?**

25 A: No, I have not previously testified before the Illinois Commerce Commission.

26 **Q: Have you previously testified before other state regulatory bodies?**

27 A: Yes, I have previously testified before the Commonwealth of Virginia State Corporation
28 Commission.¹

29 **Q: Are you sponsoring any exhibits?**

30 A: Yes, I am sponsoring the following exhibits:

- 31 • JNGO Ex. 2.01: Resume of Kyle Thomas
- 32 • JNGO Ex. 2.02: Ryan Quint et al., Practical Guidance and Considerations for Large
33 Load Interconnections, Elevate, Gridlab (May 2025)
- 34 • JNGO Ex. 2.03: Exelon Utility Companies' Transmission Facility Interconnection
35 Requirements (Jan. 1, 2024),
- 36 • JNGO Ex. 2.04: Kyle Thomas, Ryan Quint, & Jiecheng Zhao, *An Assessment of*
37 *Large Load Interconnection Risks in the Western Interconnection*, Elevate Energy
38 Consulting, (Feb. 2025)
- 39 • JNGO Ex. 2.05: North American Electric Reliability Corporation (NERC),
40 *Characteristics and Risks of Emerging Large Loads* (July 2025)
- 41 • JNGO Ex. 2.06: North American Electric Reliability Council (NERC), *Incident*
42 *Review: Considering Simultaneous Voltage-Sensitive Load Reductions* (Jan.
43 2025)
- 44 • JNGO Ex. 2.07: NERC, *Project 2020-06 Verifications of Models and Data for*
45 *Generators – Modeling Definitions* (June 2025)
- 46 • JNGO Ex. 2.08: Norris, T., Profeta, T., Patino-Echeverri, D., Cowie-Haskell, A.,
47 *Rethinking Load Growth: Assessing the Potential for Integration of Large*

¹ Virginia State Corporation Commission Case No. PUR-2020-00035.

48 *Flexible Loads in U.S. Power Systems*, Nicholas Institute for Energy,
49 Environment, and Sustainability (2025)

- 50 • JNGO Ex. 2.09: Cox, C., Schwartz, A., Stencik, D., *Bringing Data Center*
51 *Flexibility into Resource Adequacy Planning: A Case Study of NV Energy* (Sept
52 2025)

53 **Q: What is the purpose of your testimony?**

54 A: The purpose of my testimony is to evaluate ComEd's proposed tariff revisions for large
55 demand projects and highlight areas where the proposed tariff appropriately addresses large
56 load interconnection risks, and to identify gaps where the tariff is inconsistent with best
57 practices or does not sufficiently mitigate risks regarding large load interconnections.

58 **Q: What is your background with the interconnection of large loads?**

59 A: In my various roles at Dominion Energy Virginia, I was involved with the planning,
60 modeling, and studying of large load interconnections. At Elevate Energy Consulting, I am
61 involved in many industry working groups and tasks forces involving the interconnection
62 of large loads, including the NERC Large Load Task Force and the ESIG Large Load Task
63 Force, while also providing large load interconnection consulting support for various
64 clients. In partnership with GridLab, I was one of the co-authors of a recent report we
65 published in May 2025 called "Practical Guidance and Considerations for Large Load
66 Interconnections."²

² JNGO Ex. 2.02, Ryan Quint et al., Practical Guidance and Considerations for Large Load Interconnections (May 2025), Elevate, Gridlab, available at <https://gridlab.org/portfolio-item/practical-guidance-and-considerations-for-large-load-interconnections/>.

II. COMED'S PROPOSED TARIFF REVISIONS**Q: What are the main items of the ComEd filing?**

A: ComEd's filing includes the following core proposed tariff changes: (1) defines a "Large Demand Project Applicant or Customer" as a customer expecting greater than or equal to 50 MW within ten years; (2) formalizes a cluster study approach for large load interconnection requests; (3) enhances and increases initial engineering deposits; (4) proposes changes to Rider DE (distribution extension deposits and rules); and (5) proposes transmission revenue security mechanisms via a new Transmission Security Agreement (TSA) to ensure recovery of transmission costs tied to large loads.

Q: What are your takeaways of the ComEd filing?

A: ComEd recognizes the volume and systemic nature of recent and forecasted large-load requests and is moving to a cluster study practice for large load interconnections, which is consistent with industry best practices, as demonstrated in the generation interconnection context by FERC Order 2023. ComEd's proposal to require deposits and financial security for large load interconnection studies and long-lead time items are steps in the right direction to raising the barrier to the large load interconnection process. However, there are a few gaps that I have identified, and I recommend improvements/updates to address them throughout this testimony. It is important to note that some of the technical information and requirements detailed in my testimony may not be applicable to be added to the tariff itself, but may need to be added to ComEd's Facility Interconnection Requirements (FIR).³

³ JNGO Ex. 2.03, Exelon Utility Companies' Transmission Facility Interconnection Requirements (Jan. 1, 2024), available at <https://www.pjm.com/-/media/DotCom/planning/plan-standards/private-ace/eu-trans-fac-interconnection-requirements.pdf>.

That FIR document should be referenced and required by the tariff so that any large load customer is required to follow the interconnection requirements defined in the document. I do commend ComEd for proposing these tariff revisions, as it is a good start to improving the interconnections of large loads on their system. However, the additional technical improvements and recommendations provided below, I believe, are critical to holistically addressing significant bulk power system risks caused by large load interconnections. The industry has started to see early warning signs of these risks through recent data center load loss events in Virginia and Texas. While ComEd may not have seen these same issues yet as the large load growth is just beginning, getting ahead of these issues now while tariff updates and the overall large load interconnection process is being updated will help ensure better reliability for all of ComEd's customers and help ensure that risk mitigations are put in place during the interconnection process. This will reduce the risks on the system and allow ComEd to sort out costs with the large load customers during the interconnection process, rather than after the facilities are already operational and risk mitigation costs are spread to all customers.

III. COMED'S 50 MW THRESHOLD FOR LARGE LOAD PROVISIONS

Q: What are your concerns with ComEd's 50 MW threshold for large load provisions?

A: ComEd uses a fixed threshold of 50 MW within ten years to trigger the "Large Demand" provisions. Material bulk power system reliability impacts can occur from large load facilities that fall below this 50 MW threshold depending on a variety of factors, including system strength at specific locations, congestion on the system, and the type of large load facility and operational characteristics. A single MW threshold could risk missing high-impact projects that are below 50 MW in size but located in a weak, stressed, or congested

part of the transmission system. I would recommend that ComEd keep the 50 MW threshold as a primary trigger mechanism for large load provisions, but add a supplemental trigger that would invoke the large load provision/processes under a set of defined conditions, including but not limited to: (a) any project whose interconnection point is identified as “weak” or constrained in ComEd’s studies; (b) any project that requests voltage levels or in-service profiles that indicate high ramp rates, both in the MW size of ramps and the speed of the MW (fast switching of load at the second or sub-second level); or (c) any single developer proposing cumulative capacity in a geographic cluster that exceeds a defined MW threshold but within a short timeframe (months, for example). This additional supplemental trigger language when added to the tariff language would provide ComEd flexibility for unique and difficult interconnection projects.

IV. TECHNICAL CONCERNS WITH LARGE LOAD INTERCONNECTIONS

Q: What are the emerging reliability concerns associated with large loads?

A: Across the country, the electric system is experiencing unprecedented growth in large loads, including hyperscale data centers, manufacturing facilities, and industrial processes such as electrolyzers. This large load growth is dominated by one type of large load facility in particular, data centers.⁴ Unlike historical patterns of minimal to negative load growth, these new projects often involve hundreds, or even thousands, of megawatts of load seeking to interconnect within just a few years. The pace and scale of development of these large

⁴ See JNGO Ex. 2.04, Kyle Thomas, Ryan Quint, & Jiecheng Zhao, *An Assessment of Large Load Interconnection Risks in the Western Interconnection*, Elevate Energy Consulting, (Feb. 2025) (available at https://www.wecc.org/sites/default/files/documents/products/2025/Report_WECC%20Large%20Loads%20Risk%20Assessment%204.pdf).

load facilities is now far faster than the timelines for traditional transmission and distribution upgrades. In addition, the unique complex nature of these new large load facilities, especially data centers focused on the training of Artificial Intelligence systems, such as the new Large Language Models, is introducing new operational characteristics of load facilities not previously seen at this scale. These challenges are creating novel and expanding reliability risks to the bulk power system at a pace that the industry has not seen before. The NERC Large Load Task Force (LLTF) White Paper “Characteristics and Risks of Emerging Large Loads”⁵ highlights many of these risks to the bulk power system. Among the most significant include: (1) sudden, high-magnitude loss of load, which can destabilize system frequency; (2) sensitivity to voltage and frequency excursions, where large load facilities disconnect during normal protection and control functions during a fault on the bulk power system; (3) high inrush currents and fast-ramping behavior during startup or cycling that can exceed local system capabilities and drive unexpected transmission upgrades; (4) power quality impacts, including harmonic distortions, voltage flicker, and potential adverse interactions with inverter-based resources and other power electronic equipment connected to the grid. The NERC “Incident Review: Considering Simultaneous Voltage-Sensitive Load Reductions”⁶ provides concrete examples of these risks. As detailed in the report, a recent event in Virginia involved large load facilities

⁵ JNGO Ex. 2.05, North American Electric Reliability Corporation (NERC), *Characteristics and Risks of Emerging Large Loads* (July 2025) (available at https://www.nerc.com/comm/RSTCReviewItems/3_Doc_White%20Paper%20Characteristics%20and%20Risks%20of%20Emerging%20Large%20Loads.pdf).

⁶ JNGO Ex. 2.06, North American Electric Reliability Council (NERC), *Incident Review: Considering Simultaneous Voltage-Sensitive Load Reductions* (Jan. 2025) (available at https://www.nerc.com/pa/rrm/ea/Documents/Incident_Review_Large_Load_Loss.pdf).

tripping offline in large quantities (dozens of facilities), totaling approximately 1,500 MW, during a fault event that occurred on a transmission line due to a failed piece of equipment. Critically important to this event was the unexpected nature of this large load loss; the utility and grid operator were not aware that this large load loss could occur for such a fault event, as it was not modeled or captured in any studies. While this specific event did not lead to any major system disruption or outage, it is an early warning indicator that the rapid interconnection of large loads is introducing new risks to the bulk power system that must be appropriately modeled and studied during the interconnection process.

Q: What are the potential grid reliability risks when large loads connect without corresponding electricity supply commitments?

In addition to the above-mentioned reliability impacts, rapid large load growth poses a generation resource adequacy challenge. Unlike incremental demand increases that can be forecast and planned for years in advance, the sudden addition of hundreds, or even thousands, of MW of new load can create a step-change in local and regional generation capacity needs. If generation resources are not procured or available to serve these new loads, the system may face shortages during peak conditions, requiring increased reliance on imports or emergency procedures. The NERC LLTF White Paper notes that such concentrated demand can strain capacity margins and accelerate the need for new generation, which challenges the industry's existing resource planning and market activities to adequately procure the sufficient amount of generation resources. Without close coordination between utilities, who are both managing the large load growth and procuring generation resources, and ISO/RTO grid operators who are performing capacity planning and auctions, there are likely to be system-wide reliability risks in the form of

insufficient generation resource adequacy that will extend beyond the utilities' electric service and potentially impact an entire region or interconnection. I strongly recommend that ComEd work closely with PJM and other stakeholders to share large load growth and forecasts, discuss generation resource plans to meet this new load with existing or new resources, and improve generation resource adequacy studies to adequately account for a range of large load forecasts.

Q: What do you recommend that utilities and commissions do to mitigate these risks?

A: Overall, NERC has made clear that large load interconnections present emerging reliability concerns that require substantial technical considerations during the large load interconnection process. The Elevate/GridLab report provides a series of recommendations to utilities and commissions to help mitigate these bulk power system reliability risks. These recommendations include: (1) perform the full suite of models and studies of large loads during the interconnection process, including steady state, dynamics, short circuit, and electromagnetic transient studies; (2) update interconnection requirements for large loads to include mandatory ride-through capabilities, performance requirements during power system disturbances, power quality requirements, model and information sharing, and protection and control coordination activities; (3) create standardized, uniform interconnection processes for large loads with defined milestones, adequate fees for interconnections, material modification rules, and cluster study arrangements; and (4) adopt flexible interconnection arrangements for large loads. Across the country, interconnection requirements, processes, and tariffs for large loads should be updated to reflect these new requirements in order to ensure bulk power system reliability while supporting the safe, reliable, and cost-effective integration of large loads.

193 **Q: What are the issues with the technical data submittals required by ComEd?**

194 A: ComEd's testimony emphasizes detailed monthly ramp forecasts and financial
195 commitments, but it does not require the full, standardized technical data set that we
196 recommend in our report, which includes detailed load compositions of the facility, motor
197 and power-electronic details, startup profiles, operational characteristics, UPS and backup
198 generation configurations and settings, harmonic emission and other power quality
199 expectations and mitigations, protection and control settings, expected power factor, and
200 more as identified in the Elevate/GridLab report.⁷ Without this information being provided
201 to ComEd throughout the large load interconnection process, from initial application all
202 the way to commissioning and commercial operation, ComEd will be challenged with
203 adequately assessing the system reliability impacts from these large load interconnections
204 and requiring mitigating solutions to be put in place by the interconnecting customer during
205 the interconnection process. I recommend that ComEd create a tariff appendix that includes
206 a standardized data checklist, modeled after Appendix A in the Elevate/GridLab report,
207 which must accompany any large demand application. ComEd should refuse to start cluster
208 studies and system impact studies until large load applicants provide the standardized
209 technical dataset or demonstrate why certain items are not applicable. In addition, ComEd
210 should require interconnection customers to provide verified and validated models of their
211 facility to ComEd. Model Verification is defined by NERC as "the process of confirming
212 that model structure and parameter values are representative of the equipment or facility

⁷ <https://gridlab.org/wp-content/uploads/2025/07/GridLab-Report-Large-Loads-Interim-Report.pdf.zip>

213 design and settings by reviewing equipment or facility design and settings
214 documentation.”⁸ Model Validation is defined by NERC as “the process of comparing
215 simulation results with measurements to assess how closely a model’s behavior matches
216 the measured behavior.”⁹ As detailed in the Elevate/GridLab report, creating these model
217 verification and validation requirements for large load facilities is a complex process due
218 the complexity of the facilities themselves. Without having accurate models that truly
219 represent the detailed, complex dynamic characteristics of these facilities, large load
220 interconnection studies performed by ComEd will not fully capture the steady state and
221 dynamic nature of these facilities during grid disturbances and contingencies, potentially
222 leading to unidentified grid reliability risks. In addition, ComEd staff have limited time and
223 information to accurately develop models for these interconnecting facilities due to limited
224 technical knowledge of the facilities and limited access to data regarding the specific
225 equipment and configurations of the facilities. It is up to the interconnection customers
226 themselves, who know more about the facilities than anyone else could, to develop verified
227 and validated models of their large load facilities, just as is required from the developers,
228 owners, and operators of generation facilities today that interconnect into the bulk power
229 system.¹⁰ This will reduce the time and cost of the interconnection process for ComEd, and

⁸ JNGO Ex. 2.07, NERC, *Project 2020-06 Verifications of Models and Data for Generators – Modeling Definitions* (June 2025) (available at

[https://www.nerc.com/pa/Stand/Project 2020_06 Verifications of Models and Data f/2020-06%20Model%20Definitions%20Posting_Clean%20Final_June_30_2025.pdf](https://www.nerc.com/pa/Stand/Project%2020_06_Verifications_of_Models_and_Data_f/2020-06%20Model%20Definitions%20Posting_Clean%20Final_June_30_2025.pdf)).

⁹ *Id.*

¹⁰ NERC Modeling, Data, and Analysis Reliability Standards for Generators (<https://www.nerc.com/pa/Stand/Reliability%20Standards/MOD-025-2.pdf>, <https://www.nerc.com/pa/Stand/Reliability%20Standards/MOD-026-1.pdf>, <https://www.nerc.com/pa/Stand/Reliability%20Standards/MOD-027-1.pdf>, <https://www.nerc.com/pa/Stand/Reliability%20Standards/MOD-032-1.pdf>)

will improve the accuracy of the interconnection studies in identifying bulk power system risks and reliability issues so that solutions can be properly identified before the facilities go into commercial operation on the grid.

Q: What models and studies should be required during the large load interconnection process?

A: ComEd's filing memorializes cluster studies and initial engineering analyses but leaves out the specific scope details of these models and studies, such as what models to require, what studies will be performed, when are electromagnetic transient (EMT) models and studies required, and what detailed protection and control information must the interconnection customer provide to ComEd for adequate protection coordination studies. For these large load interconnections, it is essential to do the full suite of studies to adequately address bulk power system reliability issues and risks, as the industry has learned with the interconnection of generation facilities onto the system.¹¹ The complex nature of these new large load facilities, driven by incredibly fast power electronic components and protection/control systems, demand that more detailed studies are performed as a standard for each interconnection facility. Large load facilities can introduce risks that are different from what the industry has historically seen,¹² and therefore a full suite of studies is critical, including load flow, dynamic stability, protection coordination studies (including circuit breaker duty studies), and EMT studies. I recommend that ComEd update its tariff to require these specific models and studies to identify potential bulk power system reliability

¹¹ See, e.g., FERC Order 2023, *Improvements to Generator Interconnection Procedures and Agreements* (July 2023) (available at <https://www.ferc.gov/media/order-no-2023>).

¹² JNGO Ex. 2.06.

risks during the interconnection process, so that mitigations can be put in place prior to the facilities going into operations on the grid.

Q: What should be considered regarding disturbance ride-through and power quality risks from large load interconnections?

A ComEd's testimony focuses on process, deposits, and a transmission security agreement (TSA), but does not explicitly require ride-through performance metrics, harmonics limits, power factor/VAR capability, or explicit protection coordination requirements. Those technical requirements and responsibilities are captured to some extent in ComEd's Facility Interconnection Requirements document; however, that document does not expressly define what the requirements should be for large-load customers. I recommend that the tariff explicitly adopt appropriate technical requirements for large loads given the system-wide consequences of large load loss. Detailed settings and specifics of these requirements can be further defined in ComEd's FIR, but the tariff can be an overarching requirement to ensure that large load interconnecting customers provide a sufficient level of technical detail in the interconnection process. The tariff should make clear that failure to adhere to these technical requirements could result in curtailments or other types of electric service interruptions in the event risks and problems to the bulk power system are caused by an interconnecting customer. The tariff should further require that large load applicants meet the ComEd FIR document, which should be updated to include requirements for disturbance ride-through, IEEE-519 harmonic and power quality limits, VAR capability and power factor ranges, protection coordination, monitoring equipment installations, and require documented pre-commissioning ride-through tests and post-commissioning performance validation. Post-commissioning performance validation is a critical

component to include following the interconnection process. Once a large load facility becomes commercially operational, it will be interacting with the bulk power system for decades to come. Failure to require adequate monitoring of the facility could result in unmitigated bulk power system risks that impact existing customers both in terms of reliability and cost.

Q: What does it mean to have a flexible interconnection for a large load facility?

A: A flexible interconnection for a large load facility refers to an interconnection arrangement under which the facility does not receive firm, unconstrained transmission service at all times, but instead agrees to provide the utility with defined operating flexibility of the facility's loads under certain agreed upon stressed grid conditions. This flexibility can take the form of temporary curtailment of load at the facility or reliance on on-site generation resources during periods of grid stress or constraint. Various industry research, studies, and references have shown that flexibility of large load interconnections can significantly expand the amount of large load that can be interconnected without triggering immediate and costly system upgrades.^{13, 14} In practical terms, a flexible interconnection means that the large load facility contracts with the utility or grid operator to reduce or defer load when called upon by the utility/grid operator, thereby lowering system peak stress, accelerating the interconnection timeline, and reducing costs borne by all customers.

¹³ JNGO Ex. 2.08, Norris, T., Profeta, T., Patino-Echeverri, D., Cowie-Haskell, A., *Rethinking Load Growth: Assessing the Potential for Integration of Large Flexible Loads in U.S. Power Systems*, Nicholas Institute for Energy, Environment, and Sustainability (2025) (available at <https://nicholasinstitute.duke.edu/sites/default/files/publications/rethinking-load-growth.pdf>).

¹⁴ JNGO Ex. 2.09, Cox, C., Schwartz, A., Stencik, D., *Bringing Data Center Flexibility into Resource Adequacy Planning: A Case Study of NV Energy* (Sept 2025) (available at <https://gridlab.org/portfolio-item/data-center-flexibility-nv-energy-case-study-report/>).

291 **Q: Are there any considerations for a flexible interconnection framework in the**
292 **proposed tariff revisions?**

293 A: No, ComEd's filing does not create explicit tariff pathways or incentives (e.g., reduced
294 upgrade cost responsibility, expedited study priority, or contracted curtailment
295 compensation) for applicants willing to accept operational flexibility, which could include
296 interruptible service, demand response, or a customer reducing their own load and using
297 on-site customer-owned generation resources during times of peak grid stress. In our
298 report, we highlight that flexible interconnection options (firm vs. interruptible, peak
299 curtailment, flexible interconnection frameworks) can be a core tool to (a) reduce
300 immediate network upgrade needs, (b) accelerate service, and (c) lower overall costs to the
301 system when implemented with enforceable commitments. Existing processes, like the
302 Facility Interconnection Requirements document, do not address these opportunities
303 (reduced interconnection costs, increased interconnection speed, etc.) — so ComEd's tariff
304 appears to be the right vehicle to add these incentives and agreements between ComEd and
305 the large load interconnection customers. I recommend that ComEd add a new tariff section
306 to create a Flexible Interconnection Option (FIO) where an applicant may elect to take
307 service as Flexible, which defines curtailment capability, speed, dispatch rules, verified
308 performance, monitoring equipment requirements, and penalty clauses for not performing
309 flexibility rules when required, in exchange for expedited study timelines or other
310 financial/time incentives. Critically important with any FIO are two key items: (1) ensuring
311 that interconnection customers install high-resolution monitoring equipment to enable
312 performance tracking of the FIO; and (2) as part of the interconnection process, the
313 interconnecting customer must provide a detailed design of the FIO control systems and

equipment that will facilitate the actual curtailment functions of the facility—with appropriate redundancy and reliability of the system. The interconnection customer must coordinate any protection, controls, and monitoring equipment systems with the utility and grid operator to ensure appropriate visibility and overnight monitoring are possible by the utility and grid operator. Projects that choose the FIO may be prioritized, or accelerated, in the queue due to the reduced system impact under the grid’s peak stress conditions that are identified during the interconnection process by the utility and grid operator. This Flexible Interconnection Option would also enable a “Connect and Manage” approach coupled with a requirement for interconnection applicants to provide clean energy supply plans as part of their interconnection application, as recommended by JNGO Witness Saad Siddique.

V. CONCLUSION AND OVERALL RECOMMENDATIONS

Q: What should the Commission direct ComEd to do in the approved tariff changes?

A: The Commission should: (1) approve the changes to the tariff proposed by ComEd, and (2) direct ComEd to file additional tariff changes to update ComEd’s large-load interconnection study process, including at least the following elements: (a) retain the 50 MW threshold but add supplemental triggers for facilities less than 50 MW but with unique/risky interconnection locations and situations; (b) require a standardized data submittal checklist as a condition to apply to the interconnection process, including a requirement for interconnecting customers to submit validated and verified models of their facility; (c) have ComEd perform load flow, dynamic stability, EMT, and protection/short circuit studies during the new large load interconnection cluster studies; (d) explicitly require specific categories of interconnection requirements (ride-through, power quality, voltage support, power factor, protection, monitoring equipment, etc.) in the tariff as

337 binding obligations for large demand projects; (e) allow ComEd to add the specific details
338 of these studies and settings in ComEd's Facility Interconnection Requirements document;
339 (f) mandate post-commissioning monitoring and model validation with enforcement rules
340 and penalties; and (g) create a Flexible Interconnection Option with clear incentives and
341 enforceable commitments.

342 **Q: Is this necessary right now for ComEd and Illinois?**

343 A: Yes. While ComEd has yet to see some of the early warning signs of large load
344 interconnection risks that have emerged in other parts of the country, it is critical to be
345 proactive about these problems to reduce bulk power system risks and negative impacts to
346 customer rates and electric service reliability. ComEd's own testimony documents
347 unprecedented pipeline volume—more than 28,000 MW of demand applications, and
348 average new project sizes far exceeding historical norms—and acknowledges cluster
349 interactions and stranded cost risks. All of this heightens the need for more explicit
350 technical, modeling, monitoring, and flexibility provisions. The tariff should codify the
351 necessary technical and process guards before new interconnections proceed at scale.

352 **Q: Does this complete your testimony.**

353 A: Yes.